

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		$\frac{40}{T} = \cos 36^\circ$ (1) $T = 49.4$ [N] seen [1] Alternative: Resistive force = $50 \cos 36^\circ$ [1] = 40.5 [N] seen [1] Alternative: $\theta = \cos^{-1} \left(\frac{40}{50} \right)$ [1] $\theta = 36.9^\circ$ [1]	1	1		2	2	
	(b)	(i)	Horizontal component of tension increased and consequence i.e. reference to unbalanced forces (e.g. forward force > than resistive force) Accept greater horizontal resultant force [from dog]		1		1		
		(ii)	$\Sigma F = 49.4 \cos 20^\circ - 40$ [= 6.4 N] [1] $a = \frac{6.4}{35}$ ecf on ΣF [1] $a = 0.18$ [m s^{-2}] [1] N.B. If 50 N used then $\Sigma F = 7$ N and $a = 0.2$ [m s^{-2}]	1	1		3	3	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Rate of doing work = $F_H \times v$ or $F_H \times \frac{d}{t}$ [1] accept answers based on work done i.e. work = $Fx \cos \theta$ or $W = F_H x$ 1×[1] for one of: • F_H has increased [v has increased – award mark if no reference made to v or $\frac{d}{t}$ here] ✓ • $\cos \theta$ increased ✓ • as v increases drag increases, so more work is done against drag (per unit distance and time) ✓ Therefore P increased / claim is incorrect [1] accept answer based on work rather than rate of doing work			3	3		
			Question 4 total	2	4	3	9	5	0